Boosting Tomato Yields with AI-Powered Transplant Monitoring

Question 2	
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Submission time	@August 13, 2025 2:24 PM

Boosting Tomato Yields with AI-Powered Transplant Health Monitoring

Overview

Transplanting is a critical phase in tomato farming. The early weeks after moving seedlings into the field largely determine how well they establish, grow, and ultimately produce fruit. Monitoring plant health at this stage is essential—but doing so manually across large fields is both time-consuming and prone to human error.

To address this, we developed an AI-powered object detection system capable of identifying healthy and damaged tomato plants in large-scale fields. Our approach leverages aerial imagery captured by drones, stitched into a high-resolution orthomosaic for analysis. The result is a fast, accurate, and data-driven method for assessing transplant performance and guiding early interventions.

How We Did It

1. **Data Collection**

We deployed a drone to capture high-resolution images of the entire tomato field shortly after transplanting. These images were then processed into a single, seamless orthomosaic file—providing a complete view of the crop in one image.

2. Data Annotation

We took a sample of 640×640 crops from a high-resolution orthomosaic file and uploaded them to CVAT for annotations. The objects are classified into two categories: 1. Healthy, 2. Damaged.

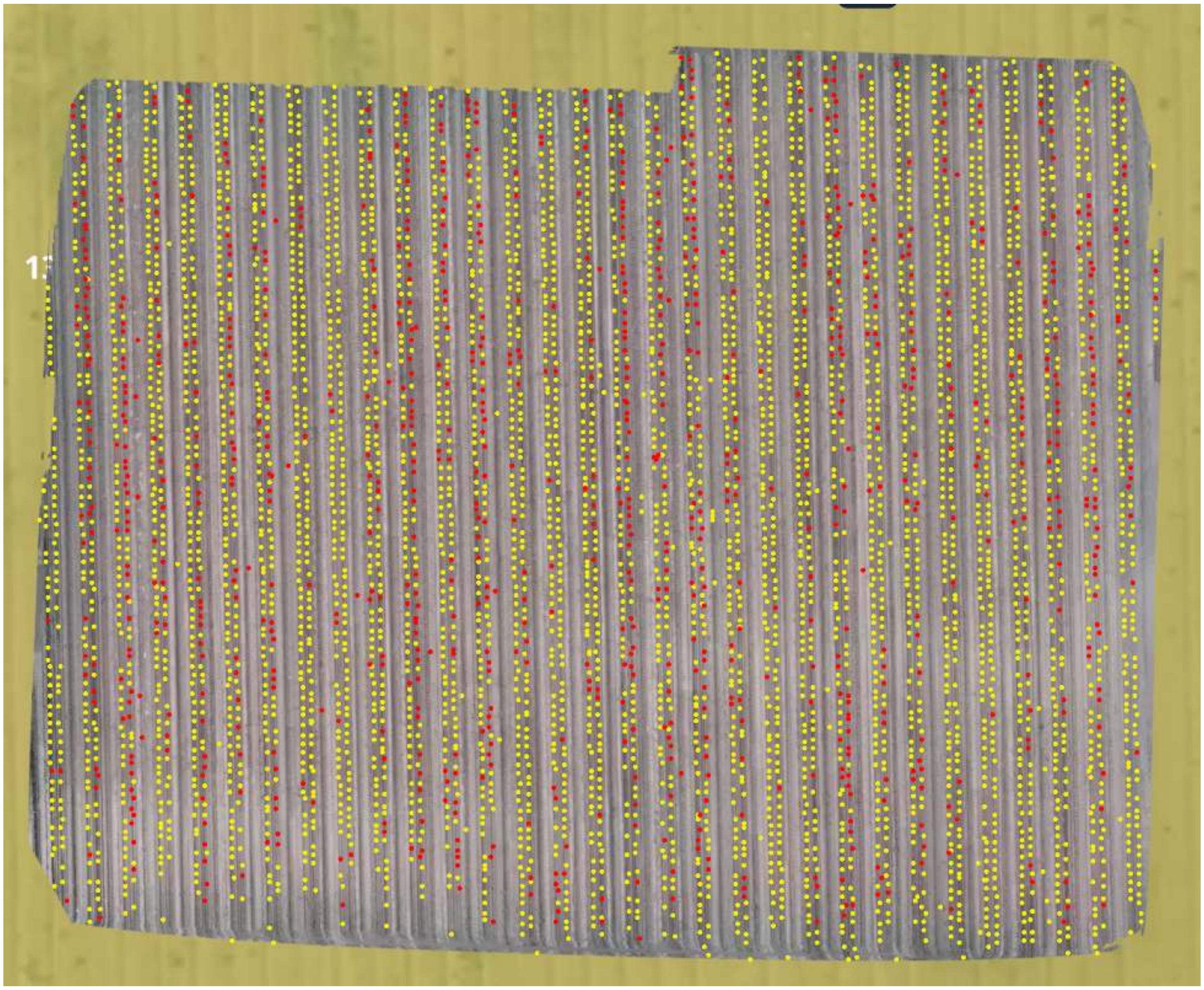
[Figure 2.1] The annotators are labeling the data

3. **Model Development**

Our AI model was trained on annotated examples of both **healthy** and **damaged** tomato plants. It uses object detection to locate each plant and classify it into one of the two categories.

3. **Inference on Orthomosaics**

Instead of analyzing each drone image individually, the stitched orthomosaic allowed us to run a single large-scale analysis, making it easier to visualize plant health across the whole field.



[Figure 3.1] shows the detections over a large othomosaic file

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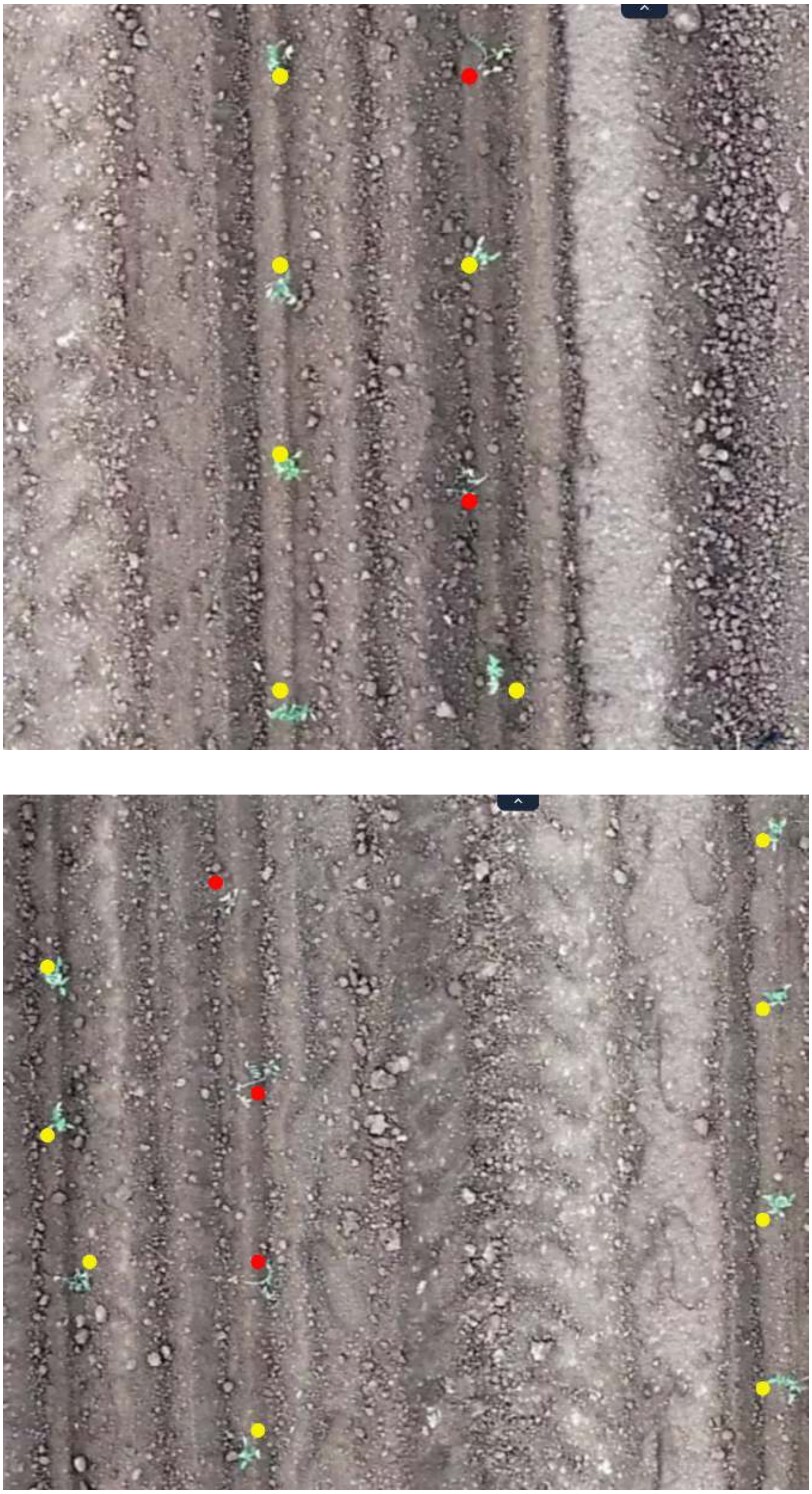
- **Yellow points** mark detected healthy plants.
- **Red points** mark detected damaged plants.

The following High-resolution close-ups highlight individual plant detections, allowing for quality checks and verification.



[Figure 3.2] shows the detections over a small area of the field

Results & Analysis

Using a confidence threshold of **0.3**, the model detected:

- **82.6% healthy plants**
- **17.4% damaged plants**

The customer's field assessment estimated **83% healthy** and **17% damaged**, meaning our results aligned very closely with ground truth. This high level of agreement indicates that the AI system can reliably replicate expert field evaluations while saving considerable time and labor.

Confidence	Total Count	Healthy	Damaged	Ratio [Healthy - Damaged]
30%	5391	4453	938	82.6% - 17.4%

Benefits of the Approach

- **Rapid Assessment:** Large fields can be analyzed in minutes once imagery is processed.
- **Objective Measurements:** Removes subjective bias from visual inspections.
- **Actionable Insights:** Clear identification of problem areas enables targeted interventions.
- **Scalability:** Works for small plots or hundreds of hectares with minimal adjustment.